

Problem Set 5

Includes problems based on assignments by David Evans and Cindy Stern from *Discrete Mathematics and its Applications* by Kenneth Rosen, from *Discrete Mathematics* by Seymour Lipschutz, and from the course textbook (MCS).

Please review and follow the instructions on Gradescope for uploading this assignment. You may use this template, or may upload a separate sheet of paper with the solutions. In either case, ensure the Collaborators, Sources, and GenAI information are included at the top of your submission. As part of the upload process on Gradescope, please make sure you indicate, as part of the upload process, which page(s) is associated with which problem.

SHOW YOUR WORK! You must show how you arrived at your answer to receive full credit.

ASSIGN PAGES! You will lose 2 points if you do not assign pages for questions.

Collaborators:

Please list any/all collaborators you worked with on the assignment:

n/a

Sources:

Please cite any additional sources you may have used:

n/a

GenAI:

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Problem 1 Proof using Primes (10 points):

Using any form of proof, prove that **if n is a prime number greater than 2, then $n + 1$ is NOT a prime number**

Useful Facts for this proof (e.g. “Given” statements). You do not need to use all of these, but may choose to do so.

- **Definition of Integer:** A whole number that has no fractional or decimal parts
- **Definition of Natural Numbers:** The set of non-negative integers $\{0, 1, 2, 3, 4, \dots\}$. The set of all Natural Numbers is often represented by $\mathbb{N}$
- **Definition of a Prime Number:** A natural number that cannot be divided by any other natural number, other than itself and 1.
- **Definition of Even:** If an integer e is even, then $e = 2k$ for some integer k
- **Definition of Odd:** If an integer o is odd, then $o = 2j + 1$ for some integer j
- **Even-Odd Lemma:** An integer is either even or odd. In other words, if an integer is NOT odd, then it is even.
- **Even-Prime Lemma:** The only even prime number is 2

#	Statement	Reason
1	n is a prime number and $n > 2$	given
2	the only even prime number is 2, and since n is a prime greater than 2, n cannot be even, by even-odd lemma, if n is not even, it must be odd	Even-Prime Lemma Even-Odd Lemma
3	$n+1 = (2j+1) + 1 = 2j+2 = 2(j+1)$	substitution
4	any integer written in $2k$ is even so $n+1$ is even	Def of Even
5	$n+1$ is even and $n+1$ is greater than 2 so $n+1$ is an even number greater than 2, it cannot be a prime number	Even-Prime Lemma

Problem 2 (6 points, 3 points each):

2.1: What is the Powerset of T , where $T = \{w, x, y, z\}$

$$P(T) = \{ "", \{w\}, \{x\}, \{y\}, \{z\}, \{w, x\}, \{w, y\}, \{w, z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{w, x, y\}, \{w, x, z\}, \{w, y, z\}, \{x, y, z\}, \{w, x, y, z\} \}$$

2.2: What is the Powerset of S , where $S = \{1, 0, \emptyset\}$ $\emptyset$

$$P(S) = \{ \emptyset, \{1\}, \{0\}, \{\emptyset\}, \{1, 0\}, \{1, \emptyset\}, \{0, \emptyset\}, \{1, 0, \emptyset\} \}$$

Problem 3 (14 points, 2 points each)

Note: Set Operations will be discussed Monday, 2/23. You may be unable to complete this problem before then

Answer the questions below based on the following sets.

$$A = \{0,1\}$$

$$B = \{0,1,2,3,4,5,6,7,8,9, a, b, c, d, e, f\}$$

$$C = \{1,2,3,5,7\}$$

$$D = \{a, b, c, d, e, f\}$$

3.1: What is the result of $A \cap C$?

$\{1\}$

3.2: What is the result of $B \cap C$?

$\{1,2,3,5,7\}$

3.3: What is the result of $A \cap D$?

$\emptyset$

3.4: What is the result of $C \cup A$?

$\{0,1,2,3,5,7\}$

3.5: What is the result of $C \cup D$?

$\{1,2,3,5,7,a,b,c,d,e,f\}$

3.6: What is the result of $B - D$?

$\{0,1,2,3,4,5,6,7,8,9\}$

3.7: What is the result of $D - B$?

$\emptyset$

Reminder:

Please assign pages, and write your collaborators, and attribute sources and GenAI use at the top of your assignment